

Task 1:

```
[postgres=# create table employees(
[postgres=# id serial,
[postgres=# first_name varchar(50),
[postgres=# last_name varchar(50),
[postgres=# department varchar(50),
[postgres=# salary numeric);
CREATE TABLE
[postgres=# select * from employees
[postgres=# ;
 id | first_name | last_name | department | salary
-----+-----+-----+-----+-----
(0 rows)

postgres=# INSERT INTO employees (first_name, last_name, department, salary)
postgres=# VALUES
postgres=# ('John', 'Smith', 'IT', 65000),
postgres=# ('Alice', 'Johnson', 'HR', 55000),
postgres=# ('Michael', 'Brown', 'Finance', 72000),
postgres=# ('Sarah', 'Davis', 'Marketing', 60000),
postgres=# ('David', 'Wilson', 'Sales', 58000);
INSERT 0 5
[postgres=# select length(first_name) from employees
[postgres=# ;
 length
-----
      4
      5
      7
      5
      5
(5 rows)
```

Task 2

```
[postgres=# create table products(id SERIAL PRIMARY KEY, product_name VARCHAR(100),  
category VARCHAR(50), price NUMERIC);  
CREATE TABLE  
postgres=# INSERT INTO products (product_name, category, price)  
postgres=# VALUES  
postgres=# ('Laptop', 'Electronics', 899.99),  
postgres=# ('Wireless Mouse', 'Electronics', 25.50),  
postgres=# ('Office Chair', 'Furniture', 150.00),  
postgres=# ('Coffee Maker', 'Appliances', 75.99),  
postgres=# ('Backpack', 'Accessories', 45.00);  
INSERT 0 5
```

Task 3

```
[postgres=# select concat(first_name, ' ', last_name) from employees;
      concat
-----
John Smith
Alice Johnson
Michael Brown
Sarah Davis
David Wilson
(5 rows)
```

Task 4

```
[postgres=# select upper(product_name) from products;  
      upper
```

```
-----  
LAPTOP  
WIRELESS MOUSE  
OFFICE CHAIR  
COFFEE MAKER  
BACKPACK  
(5 rows)
```

```
[postgres=# select lower(product_name) from products;  
      lower
```

```
-----  
laptop  
wireless mouse  
office chair  
coffee maker  
backpack  
(5 rows)
```

Task 5

```
[postgres=# select substring(department, 3) from employees;
 substring
-----
 nance
 rketing
 les
(5 rows)
```

Task 6

```
[postgres=# select lpad(id::text,3,'0') from products;
 lpad
-----
 001
 002
 003
 004
 005
(5 rows)
```

Task 7

```
[postgres=# create table sales(id SERIAL, product_name VARCHAR(100), quantity
INT, unit_price NUMERIC);
CREATE TABLE
postgres=# INSERT INTO sales (product_name, quantity, unit_price)
postgres=# VALUES
postgres=# ('Laptop', 2, 899.99),
postgres=# ('Wireless Mouse', 5, 25.50),
postgres=# ('Office Chair', 3, 150.00),
postgres=# ('Coffee Maker', 4, 75.99),
postgres=# ('Backpack', 6, 45.00);
INSERT 0 5
```

Task 8

```
[postgres=# create table expenses(id SERIAL, category VARCHAR(50), amount NUMERIC);  
CREATE TABLE  
postgres=# INSERT INTO expenses (category, amount)  
postgres=# VALUES  
postgres=# ('Rent', 1200.00),  
postgres=# ('Utilities', 250.50),  
postgres=# ('Transportation', 180.00),  
postgres=# ('Office Supplies', 95.75),  
postgres=# ('Marketing', 500.00);  
INSERT 0 5
```


Task 9

```
[postgres=# select sum(unit_price*quantity) from sales group by product_name;
 sum
-----
 127.50
 450.00
 270.00
1799.98
 303.96
(5 rows)
```

Task 10

```
[postgres=# select avg(amount) from expenses group by category;
          avg
-----
 95.7500000000000000
500.0000000000000000
250.5000000000000000
180.0000000000000000
1200.0000000000000000
(5 rows)
```

Task 11

product_name	round
Laptop	899.9900000000
Wireless Mouse	25.5000000000
Office Chair	150.0000000000
Coffee Maker	75.9900000000
Backpack	45.0000000000
(5 rows)	

Task 12

product_name	tot
Laptop	899.99
Office Chair	150.00
Coffee Maker	75.99
Backpack	45.00
Wireless Mouse	25.50
(5 rows)	

Task 13

```
[postgres=# create table orders(id SERIAL, order_date DATE, product_name VARCHAR(100),
quantity INT);
CREATE TABLE
postgres=# create table orders(id SERIAL, order_date DATE, product_name VARCHAR(100),
quantity INT);
ERROR:  relation "orders" already exists
postgres=# INSERT INTO orders (order_date, product_name, quantity)
postgres-# VALUES
postgres-# ('2026-01-05', 'Laptop', 2),
postgres-# ('2026-01-10', 'Wireless Mouse', 5),
postgres-# ('2026-02-03', 'Office Chair', 3),
postgres-# ('2026-02-15', 'Backpack', 4),
postgres-# ('2026-03-01', 'Coffee Maker', 6),
postgres-# ('2026-03-12', 'Laptop', 1),
postgres-# ('2026-03-20', 'Wireless Mouse', 8),
postgres-# ('2026-04-05', 'Office Chair', 2);
INSERT 0 8
```

Task 14

```
postgres=# select product_name, order_date - lag(order_date) over (order by order_date) as daysBetween from orders order by daysBetween desc;
```

product_name	daysbetween
Laptop	
Office Chair	24
Office Chair	16
Coffee Maker	14
Backpack	12
Laptop	11
Wireless Mouse	8
Wireless Mouse	5

(8 rows)